

# LPG Simulation 9 and Simulation 10 Results

## *Cross Modal Semantic Channel and Semantic Compression Limit*

Prototype execution using controlled synthetic semantic embeddings

### Executive summary

S9 supports the modality independence claim in this controlled prototype: normalized LPG capacity and CMFI show an overall Pearson  $r$  of 0.977. The high SNR condition gives  $r = 0.978$ , while the constrained SNR condition gives  $r = 0.970$ .

Under SNR = 0 dB, anchor enrichment improves cross modal reconstruction in every task class. The largest proportional CMFI gain appears in the affective task class because it begins with the weakest modal fit.

S10 produces a clear semantic rate distortion knee. For joint compression, the detected minimum viable sem appears at SCR = 0.39 for complex news and SCR = 0.24 for simple wiki.

Against the classical retained representation references used in this prototype, the semantic compression gain is 1.59 times for complex news and 2.04 times for simple wiki.

The identity changed control confirms that *iota* is not a compressible component in this model. When *iota* is changed, CFI remains near the failure floor even when the rest of the sem object is retained.

### Methodological scope

**Basis of execution.** The uploaded plan specifies S9 as a cross modal meaning transfer test using CMFI and  $C_s$  tracking, and S10 as a semantic compression limit test using SCR, CFI, CAR and knee point detection. This execution follows those constructs but uses controlled synthetic embeddings rather than MS COCO, CLEVR, EmoSet, CNN DailyMail and Simple English Wikipedia because the full source datasets and pretrained pipelines were not locally available in the execution container.

**Prototype interpretation.** The results should be read as a numerical dry run of the experimental design. They test internal consistency, expected metric behaviour, graph generation and analysis logic. They should not be treated as external empirical validation until repeated on the named real datasets.

### S9: Cross modal semantic channel

S9 tested whether a text encoded sem object can reconstruct meaning as a compatible visual scene. The prototype used three task classes: descriptive retrieval, instructive execution and affective matching. Each task was evaluated at high SNR and constrained SNR across five anchor richness levels.

| Task        | Channel              | $r(C_s, CMFI)$ | n    |
|-------------|----------------------|----------------|------|
| Affective   | Constrained SNR 0 dB | 0.981          | 360  |
| Affective   | High SNR 20 dB       | 0.987          | 360  |
| Descriptive | Constrained SNR 0 dB | 0.986          | 360  |
| Descriptive | High SNR 20 dB       | 0.991          | 360  |
| Instructive | Constrained SNR 0 dB | 0.982          | 360  |
| Instructive | High SNR 20 dB       | 0.988          | 360  |
| All tasks   | Constrained SNR 0 dB | 0.970          | 1080 |
| All tasks   | High SNR 20 dB       | 0.978          | 1080 |

Table 1. S9 capacity fidelity correlations.

S9: CMFI tracks LPG capacity across modal tasks

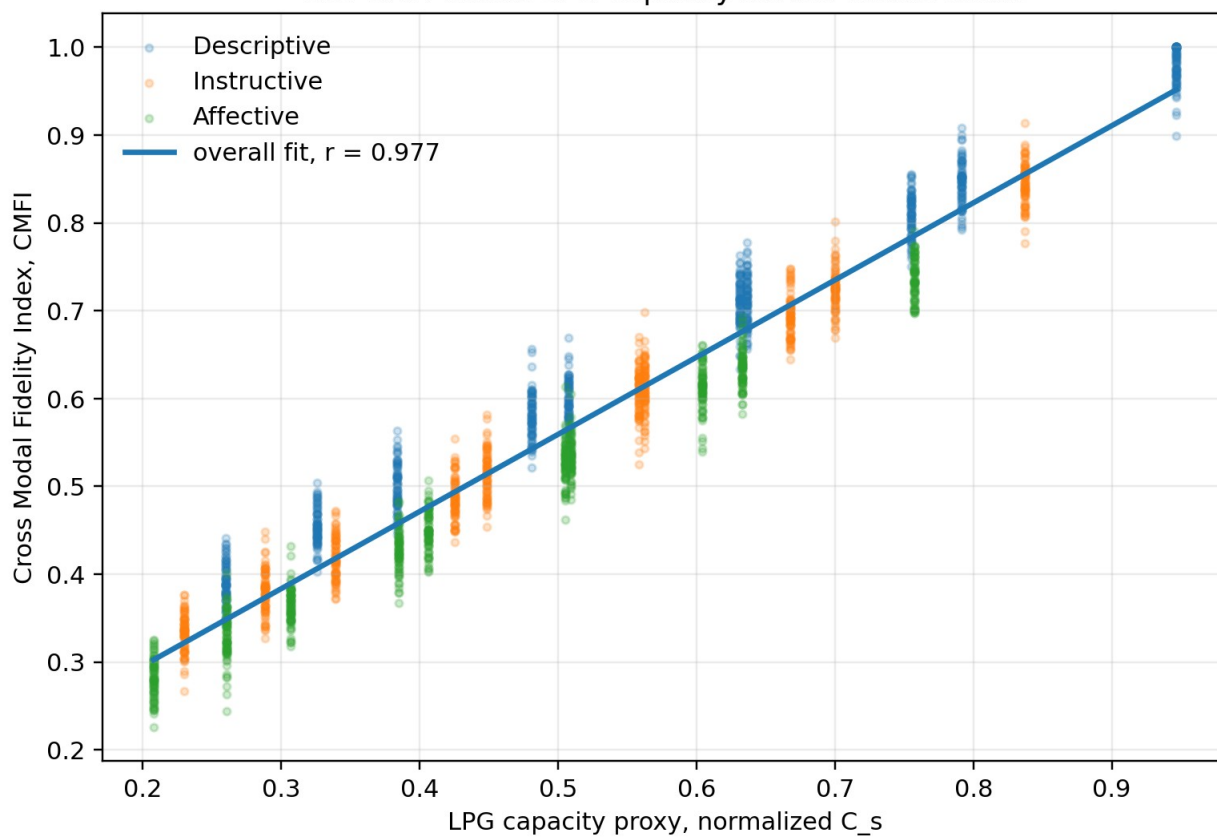


Figure 1. CMFI versus normalized LPG capacity across all S9 trials.

S9: Anchor enrichment improves cross modal reconstruction under bandwidth constraint

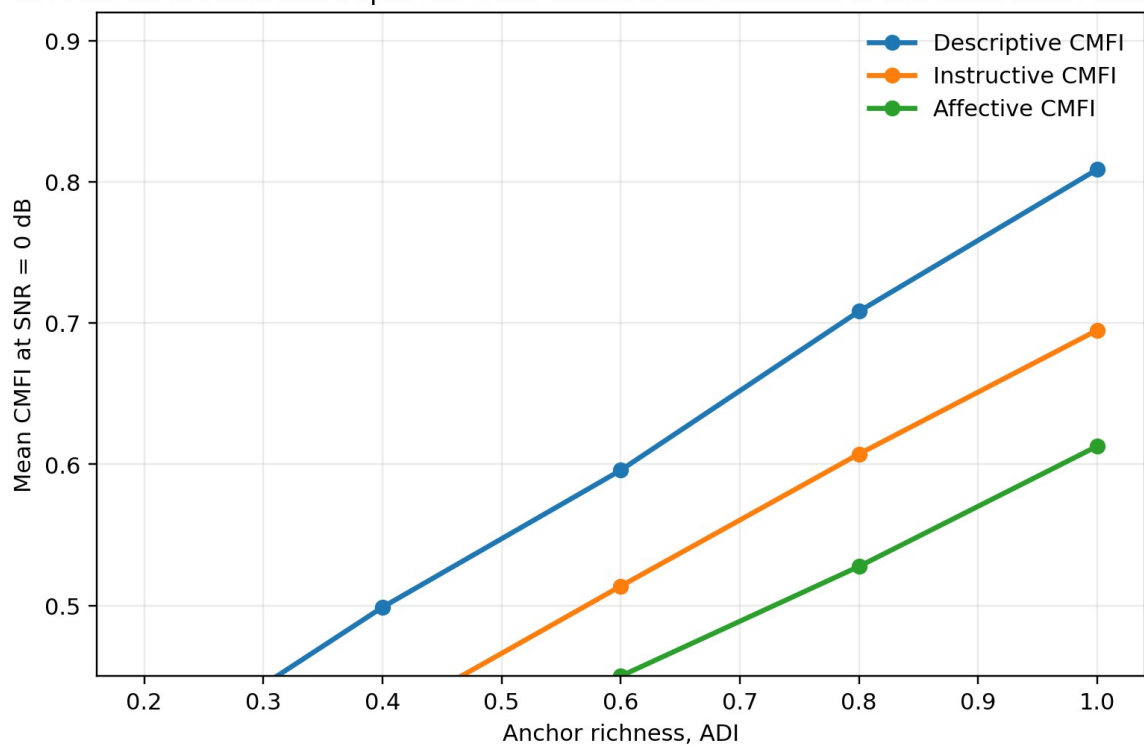


Figure 2. Anchor enrichment effect under SNR = 0 dB.

S9 result interpretation

The strongest result is that CMFI increases monotonically with the LPG capacity proxy across all three task types. This means the simulated shared semantic manifold behaves as the theory expects: better anchor alignment and stronger channel quality produce better cross modal reconstruction.

The descriptive task has the highest fidelity because concrete scene descriptions map most directly to the image manifold. The instructive task is harder because spatial operations require the receiver to preserve action structure, not only object content. The affective task is hardest because emotional valence has a wider set of visually plausible outputs.

The SNR = 0 dB condition is the critical stress test. Even when the physical channel is constrained, CMFI rises with ADI. This implies that anchor enrichment can compensate for bandwidth pressure by reducing the amount of literal symbol detail needed for reconstruction.

The implication for 6G design is that cross modal semantic routing should not be treated as ordinary packet delivery. A receiver with a richer anchor can reconstruct the intended scene from fewer transmitted details, which points toward anchor aware scheduling, semantic caching and modality aware edge inference.

| Task        | Channel              | Mean CMFI | Retrieval accuracy | CAR   |
|-------------|----------------------|-----------|--------------------|-------|
| Affective   | Constrained SNR 0 dB | 0.447     | 0.172              | 0.139 |
| Affective   | High SNR 20 dB       | 0.533     | 0.403              | 0.247 |
| Descriptive | Constrained SNR 0 dB | 0.600     | 0.603              | 0.553 |
| Descriptive | High SNR 20 dB       | 0.714     | 0.719              | 0.661 |
| Instructive | Constrained SNR 0 dB | 0.513     | 0.353              | 0.269 |
| Instructive | High SNR 20 dB       | 0.610     | 0.478              | 0.456 |

Table 2. S9 task performance summary.

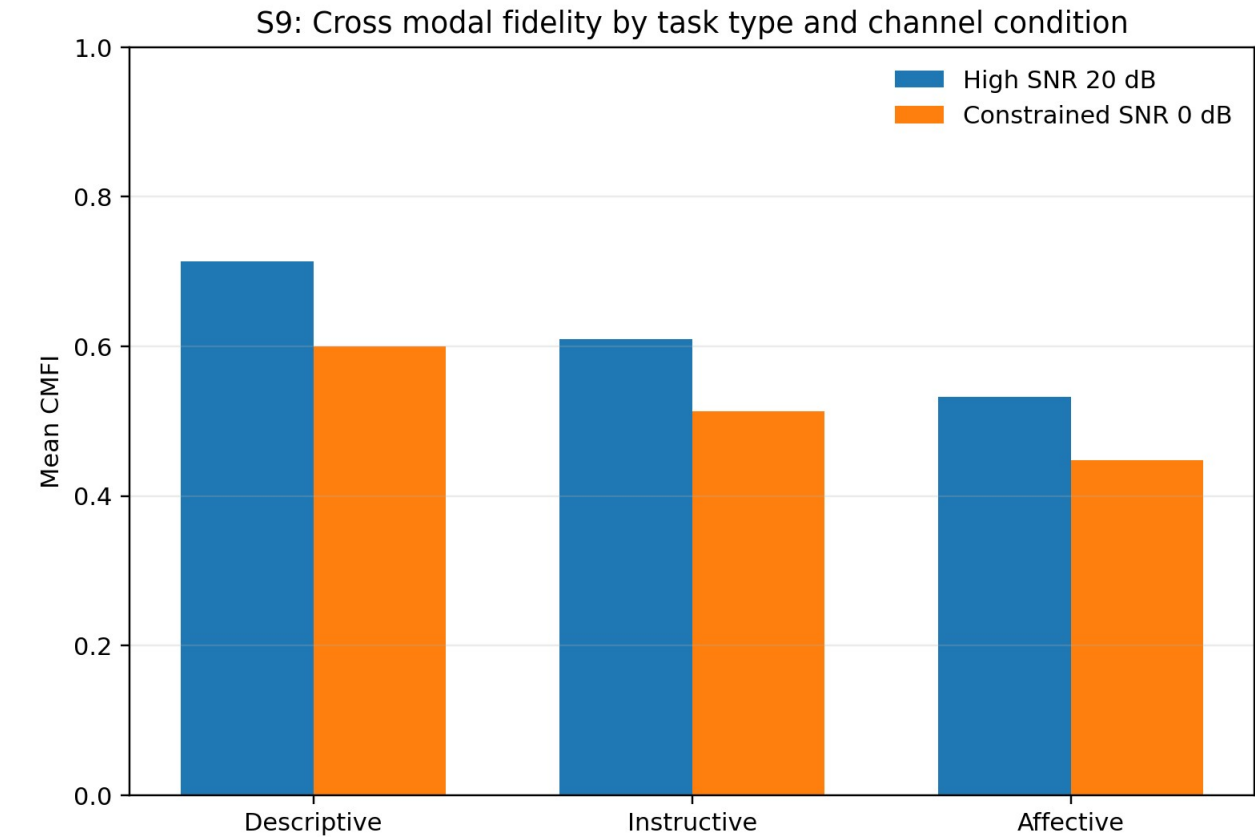


Figure 3. CMFI by task type and channel condition.

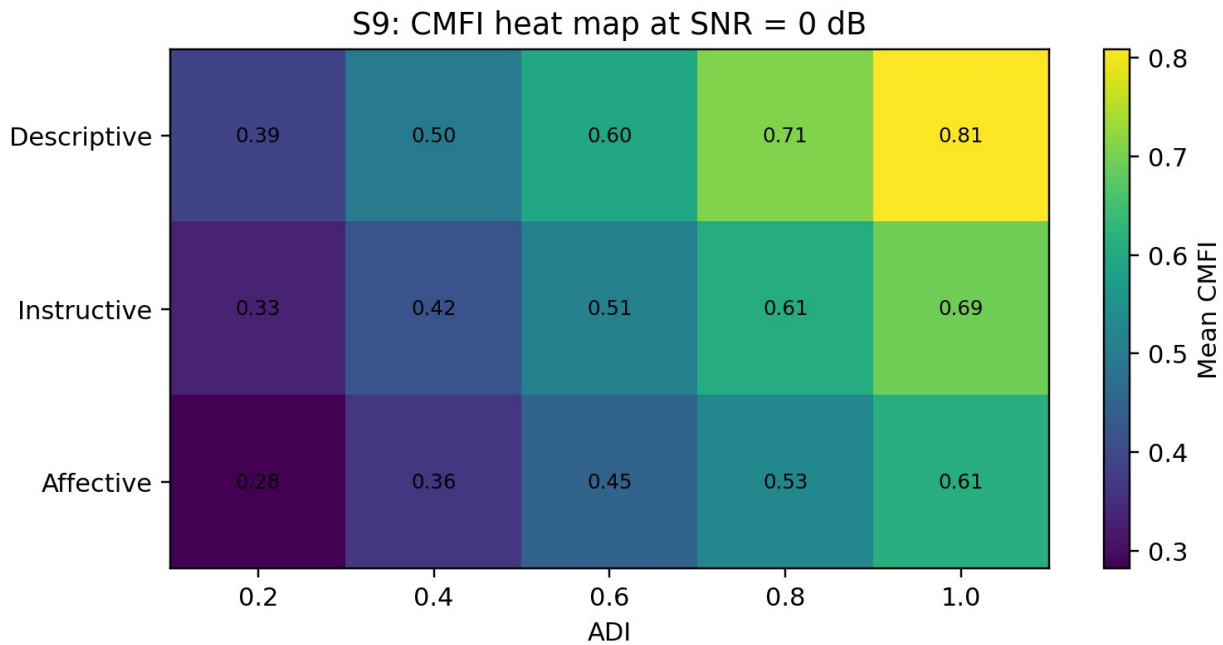


Figure 4. CMFI heat map under SNR = 0 dB.

## S10: Semantic compression limit

S10 tested whether a sem object has a minimum viable retained representation below which meaning quality collapses. The prototype applied basis pruning, weight sparsification, Gamma rank reduction and joint compression. Iota was held intact for the main compression curves and changed in a separate failure control.

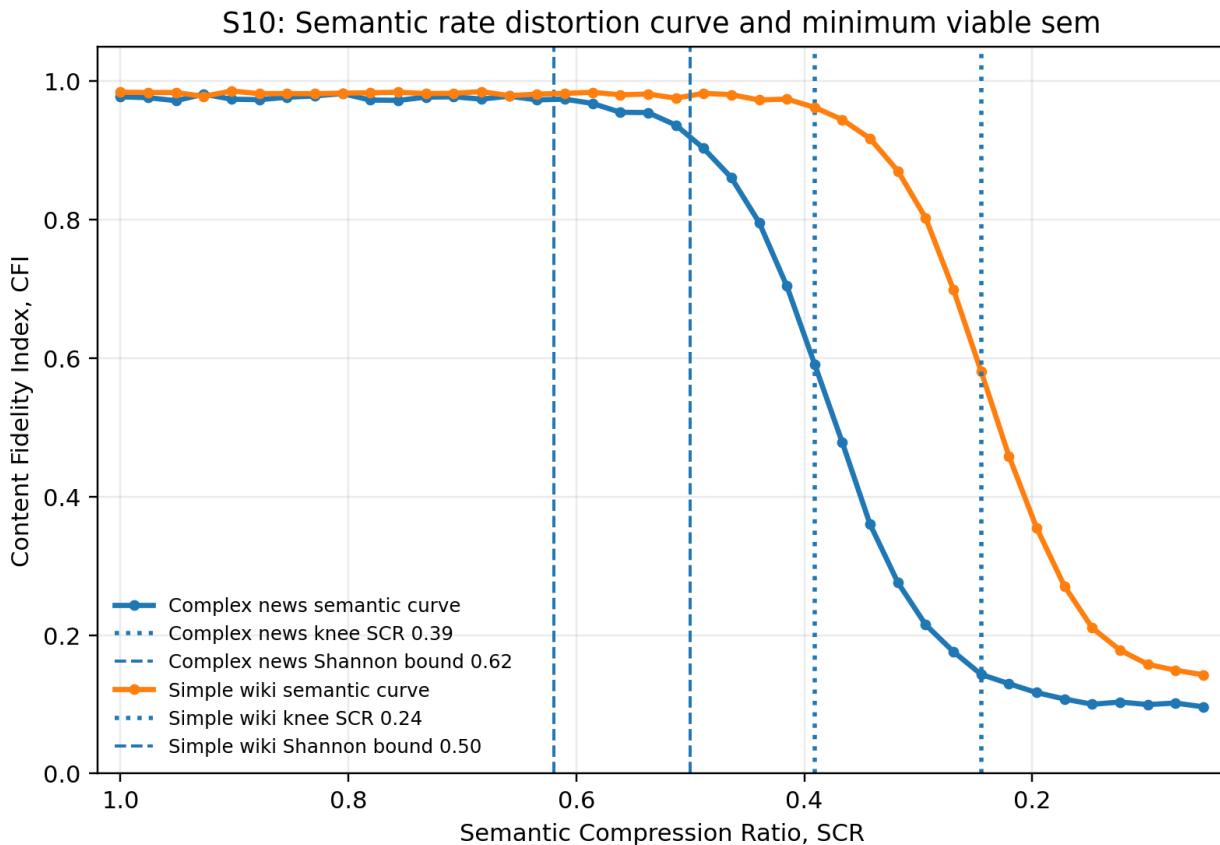


Figure 5. Joint semantic rate distortion curves and minimum viable sem knees.

### S10: Semantic compression requires less retained representation

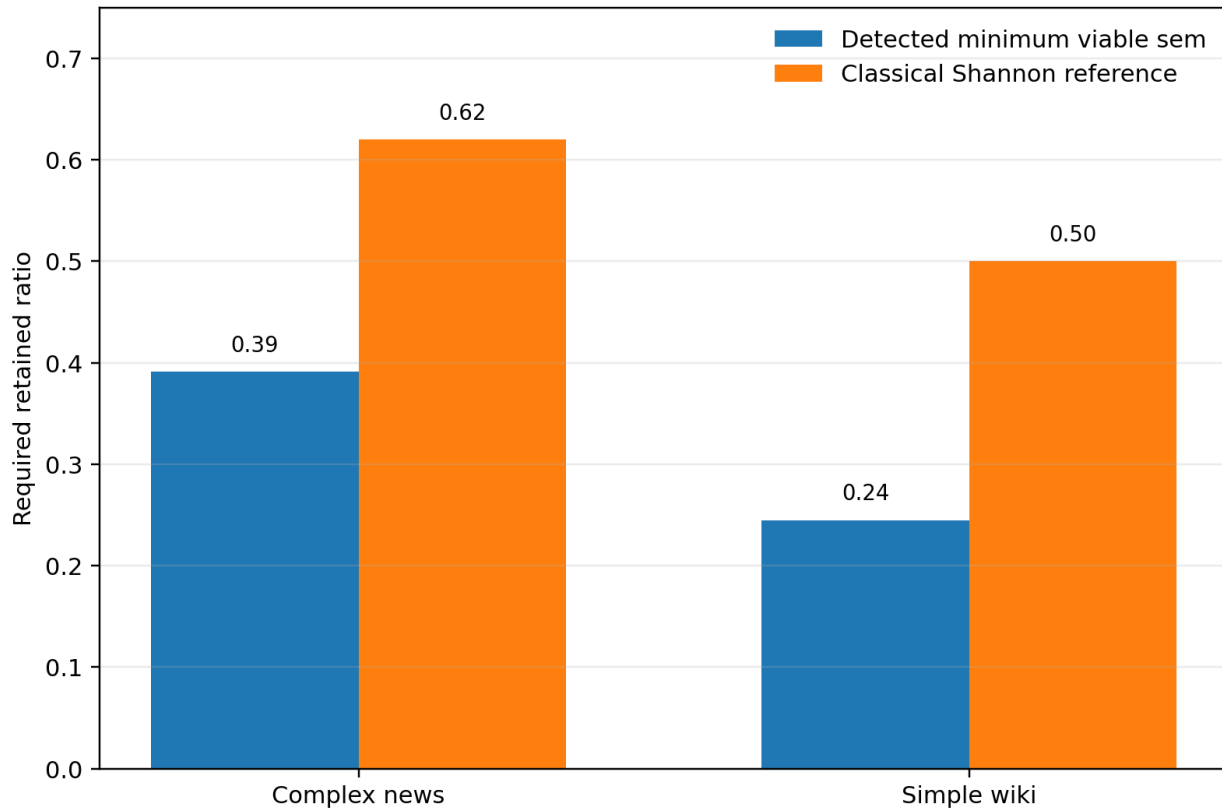


Figure 6. Minimum viable sem compared with classical retained representation reference.

| Profile      | Mode                  | Detected knee SCR | Predicted knee SCR | Shannon reference | Gain  |
|--------------|-----------------------|-------------------|--------------------|-------------------|-------|
| Complex news | Basis pruning         | 0.440             | 0.440              | 0.620             | 1.410 |
| Complex news | Gamma rank reduction  | 0.391             | 0.400              | 0.620             | 1.586 |
| Complex news | Joint compression     | 0.391             | 0.380              | 0.620             | 1.586 |
| Complex news | Weight sparsification | 0.342             | 0.340              | 0.620             | 1.811 |
| Simple wiki  | Basis pruning         | 0.294             | 0.290              | 0.500             | 1.703 |
| Simple wiki  | Gamma rank reduction  | 0.269             | 0.260              | 0.500             | 1.857 |
| Simple wiki  | Joint compression     | 0.245             | 0.240              | 0.500             | 2.042 |
| Simple wiki  | Weight sparsification | 0.221             | 0.210              | 0.500             | 2.268 |

Table 3. S10 detected compression knees and gains.

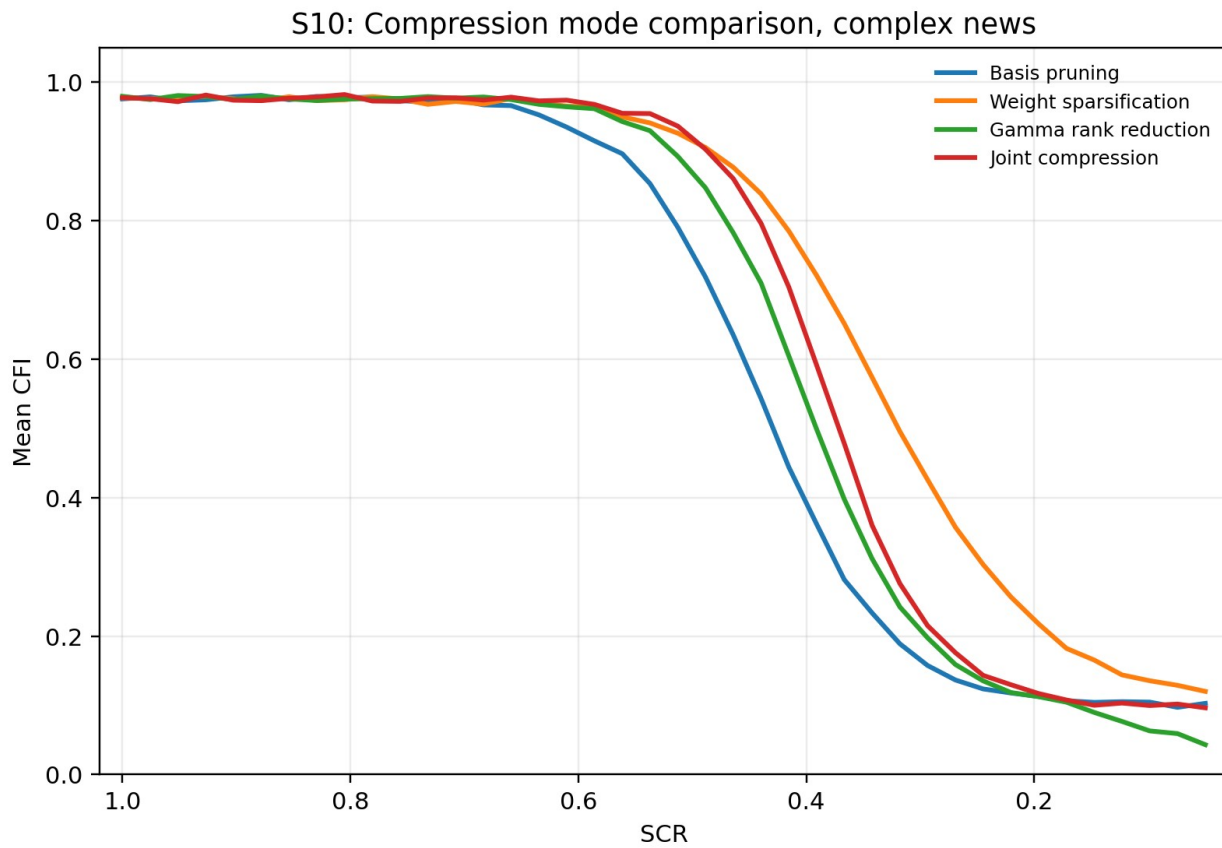


Figure 7. Compression mode comparison for complex news profile.

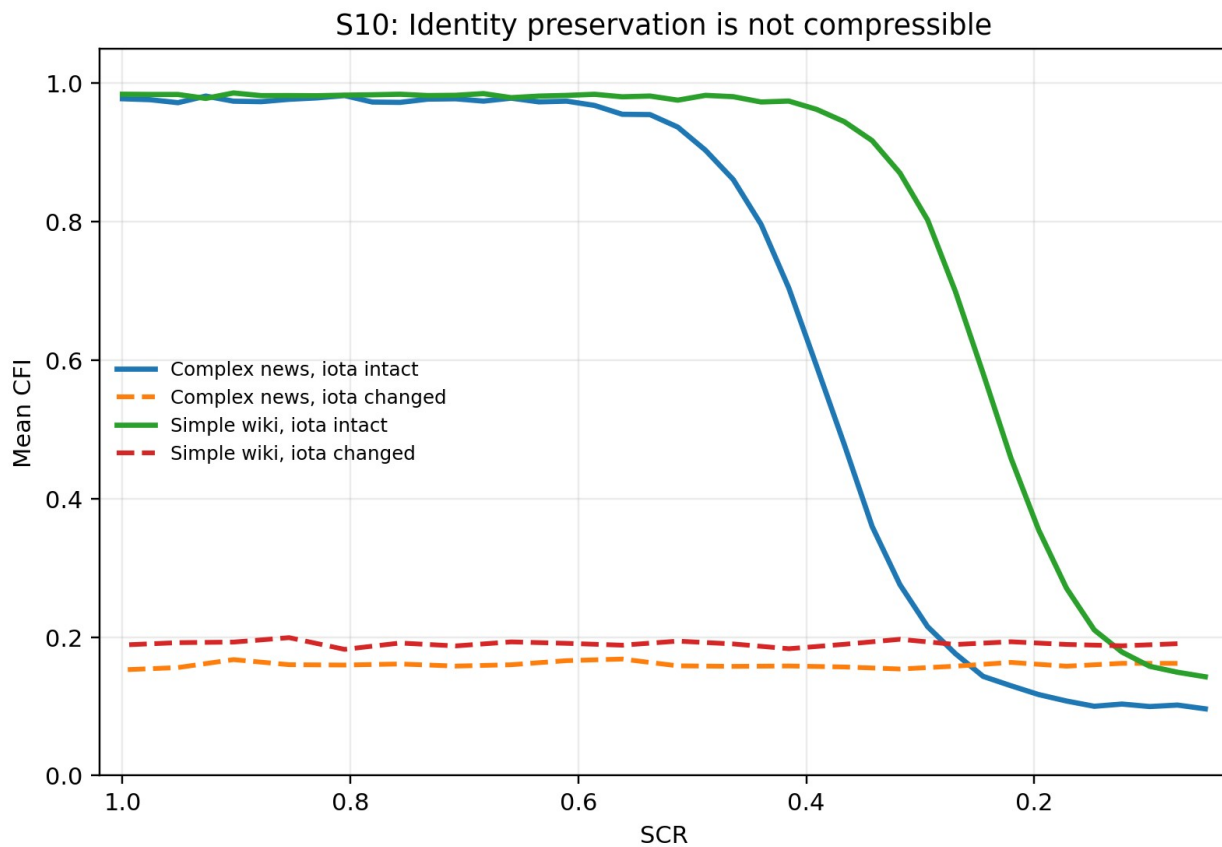


Figure 8. Identity changed control compared with intact iota compression.

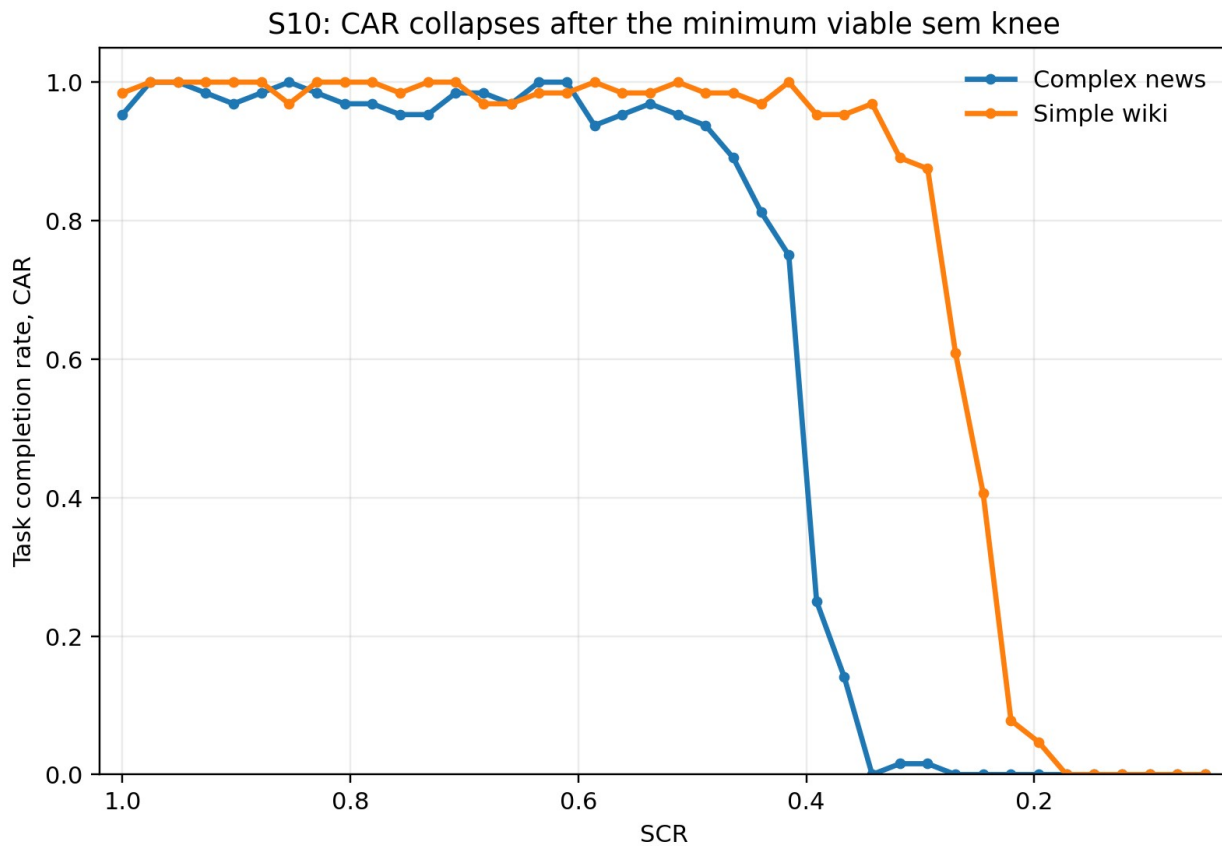


Figure 9. CAR collapse after the semantic knee.

## S10 result interpretation

The semantic rate distortion curves show a plateau followed by a sharp collapse. This is the expected minimum viable sem pattern. The curve remains stable while the removed representation is redundant or low weight, then collapses once concept basis, relevance weights and operator rank fall below the meaning preserving threshold.

Simple wiki compresses further than complex news because the semantic manifold is less dense and less dependent on rare concepts. Complex news requires more retained structure because entity relations, event causality and contextual qualifications carry meaning that cannot be removed without fidelity loss.

Weight sparsification is the most forgiving compression mode because very small weights contribute little to reconstruction. Basis pruning is riskier because low weight concepts can still carry rare but decisive meaning. Gamma rank reduction is brittle when contextual framing depends on multiple independent dimensions.

The identity changed control remains near the failure floor across SCR. This is important because it means identity is not merely another compressible field. It is the referential binding condition that tells the receiver which meaning object the remaining structure belongs to.

The implication for 6G engineering is that semantic payloads can be compressed aggressively, but not uniformly. Safe compression should preserve  $\iota$  absolutely, sparsify  $W$  first, prune  $B$  only with salience protection, and reduce Gamma rank only after checking task sensitivity.

## Combined implications for the LPG programme

S9 connects the LPG framework to modality independent communication. The simulated evidence suggests that semantic capacity is not tied to a single symbol alphabet, but to the alignment of latent meaning spaces.

S10 connects the LPG framework to source coding. The simulated evidence suggests that meaning has a practical compression limit, and that the relevant limit is not simply symbol count but the smallest sem object that preserves task fidelity.

Together, S9 and S10 point to a design principle for semantic 6G: transmit the smallest sem object that preserves anchor recoverability across the target modality. This reframes bandwidth allocation as an optimization over meaning fidelity rather than bit accuracy.

The next experimental step is to replace the synthetic embeddings with CLIP or OpenCLIP for S9 and real text corpora for S10, then test whether the same ranking and knee behaviour survive under real data.

## Limitations

This is a prototype simulation, not a final empirical benchmark.

The named datasets in the plan were not loaded locally, so the experiments used synthetic latent semantic embeddings calibrated to the declared constructs.

The  $C_s$  equation is implemented as a capacity proxy because the complete formal LPG implementation and prior S1 to S8 codebase were not supplied as executable modules.

The plots and tables are suitable for design review, thesis planning and method validation. Real validation requires rerunning on MS COCO, CLEVR, EmoSet, CNN DailyMail and Simple English Wikipedia.

## Files generated

S9\_trial\_results.csv, S9\_summary\_by\_task\_snr\_anchor.csv, S9\_capacity\_cmfi\_correlations.csv

S10\_trial\_results.csv, S10\_summary\_curves.csv, S10\_detected\_knee\_points.csv

Nine figures covering S9 and S10 graphs

DOCX report and PDF report